

Telecom Customer Churn Analysis Using SQL & Power BI

Executive Project Report

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Role: Data Analyst

Tools: MySQL • SQL • Power BI • Microsoft Excel

Dataset: Telecom Customer Churn Dataset (Kaggle)

Executive Summary

Customer retention remains one of the most significant drivers of long-term profitability in the telecommunications industry. Acquiring new customers is substantially more expensive than retaining existing ones, making churn analysis a critical business function.

This project presents an end-to-end analysis of a telecom customer dataset containing **7,043 customer records**. Using SQL for data cleaning, transformation, exploratory analysis, and advanced querying, alongside Power BI for visualization, the project identifies the primary factors influencing customer churn and provides actionable recommendations to improve customer retention and maximize customer lifetime value.

The analysis found that **26.54% of customers had churned**, with the highest risk concentrated among customers on month-to-month contracts, electronic check payment users, customers without Online Security or Tech Support, and customers within their first twelve months of service.

Business Problem

The telecom company is experiencing customer attrition that negatively affects recurring revenue and long-term profitability.

The business seeks to answer the following questions:

- What is the current customer churn rate?
- Which customer segments are most likely to churn?
- Which products or services influence customer retention?
- Which customers generate the greatest revenue?
- What actions can reduce customer churn?

The objective of this project is to transform raw customer data into strategic insights that support evidence-based business decisions.

Dataset Overview

The analysis was conducted using the Telecom Customer Churn dataset obtained from Kaggle.

Dataset Statistics

Metric	Value
Total Customers	7,043
Total Features	21
Active Customers	5,174
Churned Customers	1,869
Overall Churn Rate	26.54%

Key Variables

- Customer ID
- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure
- Contract Type
- Internet Service
- Payment Method
- Monthly Charges
- Total Charges
- Churn Status

SQL Workflow

A structured SQL workflow was followed to ensure data quality and reliable business insights.

Phase 1 — Database Setup

- Created a dedicated MySQL database.
- Imported the telecom dataset.
- Defined appropriate data types.

- Established Customer ID as the primary key.

Phase 2 — Data Inspection

Performed data validation by:

- Checking total records
- Inspecting table structure
- Verifying unique customer IDs
- Identifying missing values
- Detecting duplicate records
- Reviewing categorical variables

Phase 3 — Data Cleaning

Improved data quality by:

- Converting blank TotalCharges values to NULL
- Changing TotalCharges from text to numeric format
- Validating numeric fields
- Confirming data consistency
- Ensuring analysis-ready data

Phase 4 — Exploratory Data Analysis (EDA)

Analyzed customer characteristics across:

- Customer demographics
- Service adoption
- Revenue generation
- Customer contracts
- Payment methods
- Customer tenure

Phase 5 — Advanced SQL Analysis

Applied advanced SQL techniques including:

- Common Table Expressions (CTEs)
 - Window Functions
 - CASE Statements
 - Subqueries
 - Aggregate Functions
 - Ranking Functions
 - Customer Segmentation
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Data Cleaning

Before conducting analysis, the dataset underwent a comprehensive quality assessment.

Data quality checks included:

- Missing value detection
- Blank value identification
- Duplicate record validation
- Data type verification
- Primary key validation
- Categorical consistency checks

A notable issue involved blank values within the **TotalCharges** column. These records corresponded to customers with zero tenure and were converted to NULL before the column was transformed into a decimal data type for accurate revenue analysis.

Following the cleaning process, the dataset was confirmed to be complete, consistent, and suitable for analysis.

Exploratory Analysis

The exploratory analysis focused on understanding customer behavior, service usage, revenue patterns, and churn characteristics.

The investigation was organized into five business-focused sections:

1. Customer Overview

- Total Customers
- Active Customers
- Churned Customers
- Overall Churn Rate

2. Customer Demographics

- Gender
- Senior Citizens
- Partner Status
- Dependents

3. Customer Services

- Internet Service
- Online Security

- Online Backup
- Device Protection
- Tech Support
- Streaming Services

4. Revenue Analysis

- Total Revenue
- Revenue by Contract
- Revenue by Internet Service
- Revenue by Payment Method
- Customer Lifetime Value

5. Advanced Churn Analysis

- Contract Risk
 - Payment Risk
 - Tenure Risk
 - Customer Ranking
 - Revenue Ranking
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Key Findings

The analysis produced several high-impact business insights.

Customer Retention

- Overall churn rate stands at **26.54%**.
- Nearly one out of every four customers has discontinued service.

Contract Type

- Month-to-month contracts exhibit the highest churn rate (**42.71%**).
- Two-year contracts have the lowest churn rate (**2.83%**), highlighting the stabilizing effect of long-term agreements.

Internet Service

- Fiber Optic customers experience the highest churn rate (**41.89%**).
- Customers without internet service show minimal churn (**7.40%**).

Customer Support Services

Customers without value-added services are significantly more likely to churn:

- Online Security: **41.77%**

- Tech Support: **41.64%**
- Online Backup: **39.93%**
- Device Protection: **39.13%**

Payment Method

Electronic Check customers show the highest churn rate (**45.29%**), while customers using automatic payment methods exhibit substantially lower churn.

Customer Tenure

Customers with less than one year of tenure have the highest churn rate (**47.44%**), indicating that customer retention efforts should be concentrated during the onboarding period.

Revenue

- Total Revenue Generated: **16.06 Million**
- Two-Year Contract customers contribute the highest average customer value.
- Retained customers generate substantially higher lifetime revenue than churned customers.

Business Recommendations

Based on the findings, the following strategic actions are recommended:

1. Encourage Long-Term Contracts

Introduce discounts, loyalty rewards, or promotional incentives that encourage customers to transition from month-to-month plans to annual or multi-year contracts.

2. Improve Early Customer Retention

Develop proactive onboarding and engagement programs targeting customers within their first twelve months of service.

3. Promote Automatic Payment Methods

Encourage customers to migrate from Electronic Check to automatic payment methods through convenience incentives or discounts.

4. Bundle Value-Added Services

Increase adoption of Online Security, Tech Support, Online Backup, and Device Protection by offering bundled service packages.

5. Prioritize High-Risk Customers

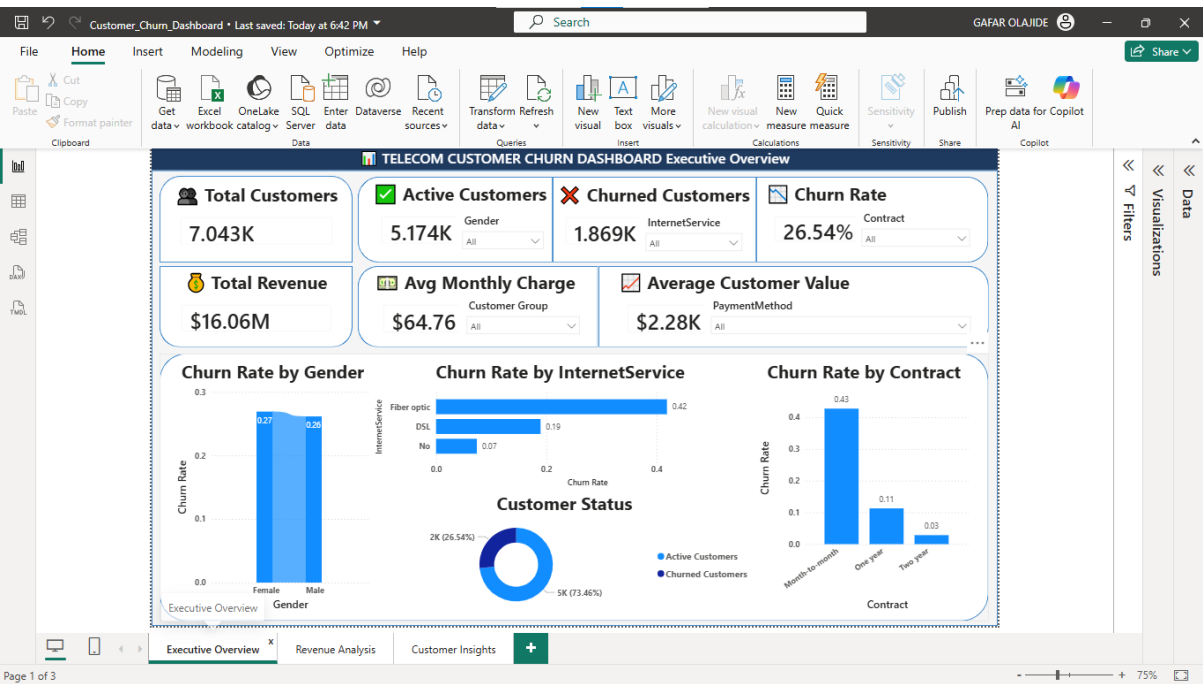
Implement predictive retention campaigns focused on customer segments exhibiting the highest churn probabilities.

Dashboard Overview

An interactive Power BI dashboard was developed to communicate insights to decision-makers.

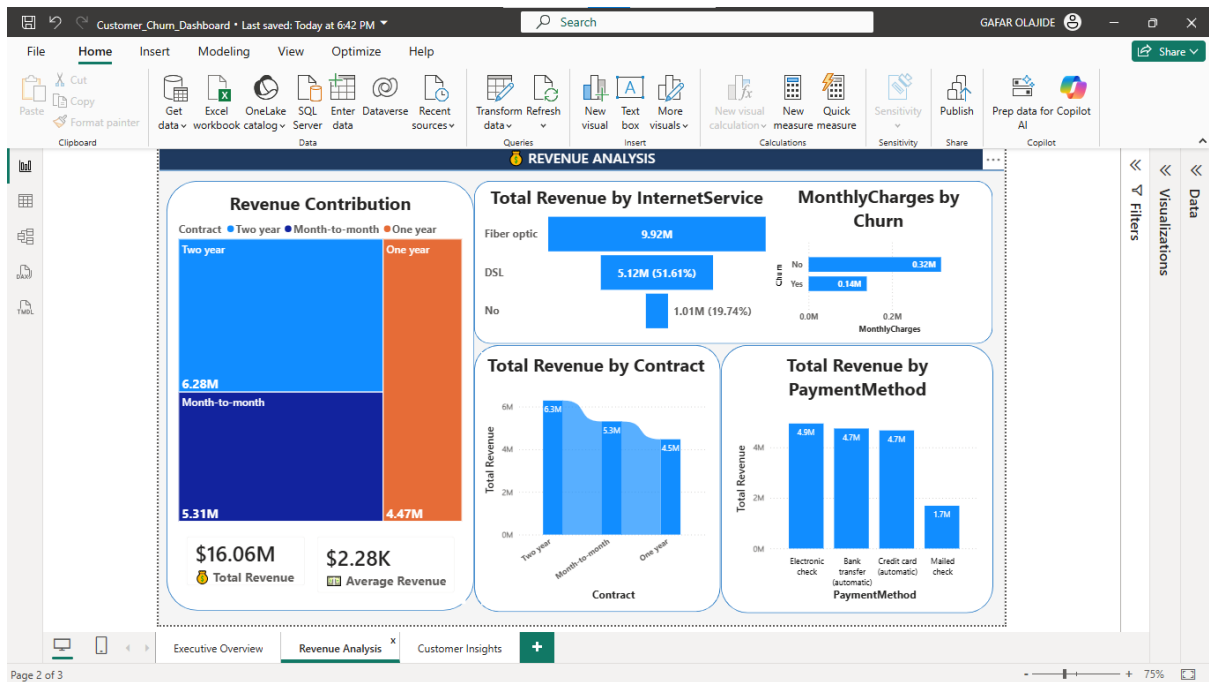
The dashboard consists of three pages:

Executive Overview



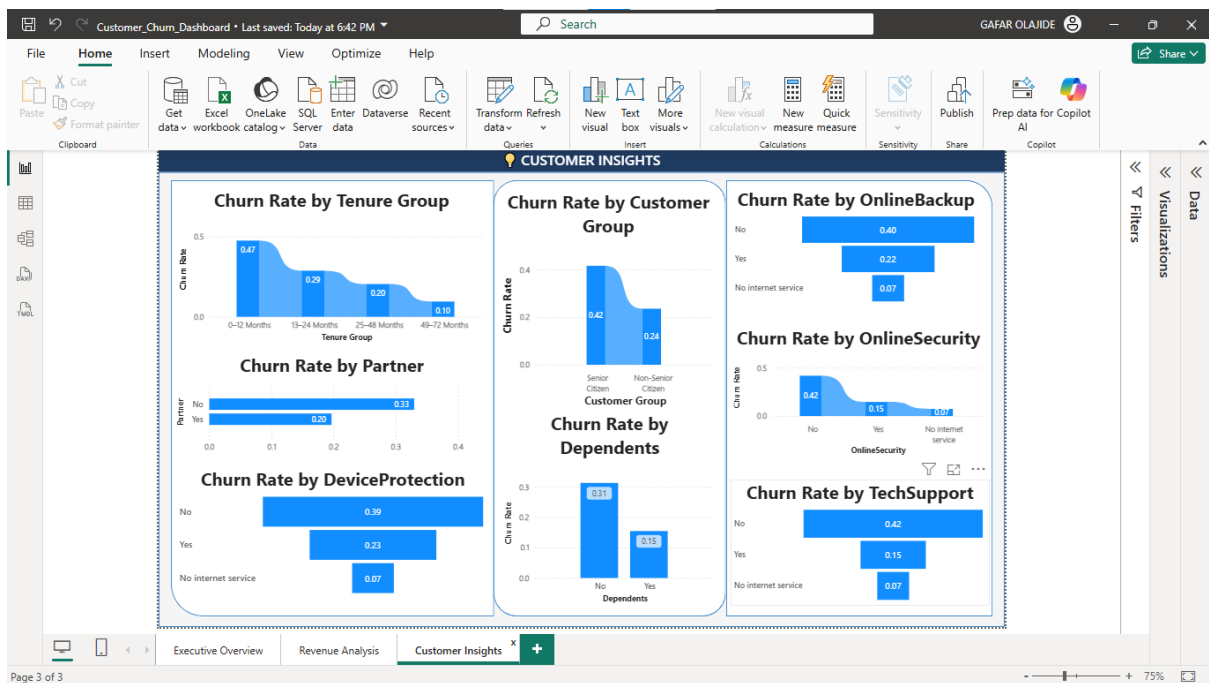
Provides high-level KPIs, customer distribution, churn overview, and contract performance.

Revenue Analysis



Examines revenue performance by contract type, payment method, and internet service while highlighting customer lifetime value.

Customer Insights



Explores churn behavior across customer demographics, tenure groups, service adoption, and support features.

The dashboard enables business users to interactively filter data and investigate churn patterns across multiple customer segments.

Conclusion

This project demonstrates how SQL and Power BI can be combined to transform raw customer data into actionable business intelligence.

Through comprehensive data preparation, exploratory analysis, advanced SQL querying, and interactive dashboard development, the analysis identified the key drivers of customer churn and proposed practical strategies to improve customer retention and increase long-term revenue.

Beyond the technical implementation, the project emphasizes translating analytical findings into business decisions—an essential capability for modern Data Analysts and Business Intelligence professionals.